Therapeutic Peptides in Rheumatoid and Osteoarthritis: The Dream of Achieving More Tolerable Therapies, or Not?

Abstract

This article presents information from studies, trials, and the literature about chronic inflammatory diseases such as rheumatoid and osteoarthritis, as well as about peptides and their uses in a therapeutic setting, in this case, in arthritic diseases. Some of the peptides explored in this article have a direct effect on the inhibition of inflammation, others enhance cartilage rebuilding, or they can help and improve existing treatments' effects whether through easing the delivery of the drug to the desired site or exacerbating its effects.

Introduction

Inflammation is a vital immune response, which could be tied to tissue malfunction and a loss of homeostatic imbalance of one or several physiological systems. It could be triggered by pattern recognition receptors (PRRs), recognizing pathogen-associated molecular patterns (PAMPs), or danger-associated molecular patterns (DAMPs). Moreover, cytokines and chemokines can stimulate cellular inflammatory processes not only by increasing the permeability of the vascular system, but by also activating the endothelial system, thus drawing more immune cells to the site of infection is done under the effect of chemokines. Additionally, prostaglandins are responsible for the signs and symptoms such as fever and weariness.(1)

Generally, inflammation can be classified as acute or chronic. There are myriads of diseases caused by chronic inflammation, and of them would be rheumatoid and osteoarthritis.

Rheumatoid arthritis

Rheumatoid arthritis is a multi-factorial chronic joint tissue inflammation disease, mostly affects small joints and in advanced cases some internal organs could be affected. Autoantibodies produced by plasma cells are detected before any sign of inflammation. In RA patients, the synovial tissue becomes either ectopic lymphoid structures (ELS) or tertiary lymphoid tissue (TLT). There are two important pathogenetic changes in their synovial membrane, first macrophage-like synoviocytes (MLSs) and fibroblast like synoviocytes (FLSs)—are increased and activated, resulting in a significant expansion of the intima. The adaptive immune cell infiltrates into the synovial sub-lining which leads to the development of a recognizable "pannus" at the cartilage-bone contacts which leads to damage later in the disease course. CD4+ T cells play major role by creating ectopic germ centers, while B cells, plasma blasts, and plasma cells produce rheumatoid factors (RFs) or ACPAs. IgA is the most commonly produced antibody in RA. Data has shown that IgA-ACPAs were increased and highly specific in the early clinical and preclinical stages of RA.(1)

Studies and clinical trials have been developed dedicated to finding and exploring different types of arthritis therapies, and lately, peptides have been a new area that scientists have been trying to tackle.

Therapeutic Peptides

Peptides are small sequences of amino acids that with less than 50 residues combine the traits of large proteins and small molecule drugs. (2) This allows peptides to act with higher selectivity and specificity thus leading to a lesser chance of off-target adverse reactions (2) and drug-drug interactions (3). Moreover, they have a lesser likelihood of systematic toxicity since the products of their break down are amino acids.(2)

Peptides use in RA

Studies and trials demonstrated different ways peptides could be used in the management of arthritis. For example, food-derived peptides exert important activity such as antioxidant and anti-inflammatory effects. For RA, its main acting mechanism can be correlated to oxidative stress and inflammation.

A paper mentions peptides sourced from plants like Lunasin was found to reduce the pro-inflammatory cytokines and help in repairing the cartilage. Additionally, chicken FDPs can improve the integrity of bones and prevent excessive formation in cartilage, as well as decreasing the pro-inflammatory cytokines in rat experimental models. In rabbit experimental models, chicken leg extract digest facilitates the cartilage matrix and increase production of Acid mucopolysaccharide.

Similarly, bovine FDPs showed promising results, as they reduced the cartilage degeneration in rat experimental models, reduced synovial hyperplasia and increased the number of cartilage cells and proteoglycan matrix in mice experimental models. They were also found to enhance hyaline cartilage while preventing fibrous tissue formation in chondrocytes, as well as inducing type II collagen and inhibiting type-I collagen deposition.

In addition to terrestrial resources, deep-water ocean fish skin is rich in collagen. When administered to arthritic dogs, it was found that the levels of MMP-3 were reduced, and it improved their pain and lameness. It also increases types I and II collagen protein synthesis mechanism while suppressing both transcription and expression of inflammatory proteases including ADAMTS5, HTRA1 and Cox-2.

Furthermore, collagen peptides from Pangasius hypophthalmus's skin also reduced inflammation by increasing the expression of the genes of aggrecan and collagen type II alpha chain. They suppressed tibial subchondral bone microstructural damage, proteoglycan loss reduction, and upregulation and downregulation of type II collagen and MMP-13, respectively in the cartilaginous tissues after oral administration to rabbits.

Finally, collagen peptides derived from Walleye pollock skin prevented articular cartilage damage while decreasing sera NO and malondialdehyde levels.(1)

On another note, a different study points out that the problem with modern RA treatments such as (Disease-Modifying Antirheumatic Drugs “DMARDs”), which is that they block TNF-alpha after it’s produced which makes them a temporary treatment, but using peptides prevents TNF secretion from macrophages and at the same time, it doesn’t affect lymphocyte that helps us in fighting infections, which leads to decreased side effects.

In Vivo experiments show that linear combined peptides show more stability and biological activity that contradicts with single amino acids, but in order to treat RA it has to be in low dosage because high dosages cause off-target effects and cause stability in blood and correct 3D orientation. While maintaining the same effectiveness as currently used RA treatments such as (Humiral/Enbrel), peptides showed stronger signaling effects more targeted and potentially safer that can be used as a main treatment with lesser side effects.(4)

Peptides and Delivery Systems

Peptides were also found to aid in the mobilization of RA drugs. A study was conducted to create a better more affective way for treatment transportation, in which liposomal nanoparticles were used to entrap the treatment, thus leading to an improvement in the bioavailability and the delivery of poorly soluble treatments, as well as an increase in the half-life of the drug. Using peptide ART-1 displaying liposomes with entrapped IL-27 was more impactful on inhibiting arthritis than with liposomes without the displayed peptide in rat adjuvant arthritis models. Moreover, the paper measured the effect of peptides on the Zeta potential (ZP), which is the overall charge on the nanoparticles, and it was demonstrated that ART-1 had a pronounced effect on the ZP. The ZP induces the repulsion of nanoparticles, thus preventing aggregation and in turn, enhancing physical stability in vivo/in vitro. (5)

Another study focuses on improving treatment for rheumatoid arthritis (RA), using a peptide guided drug delivery strategy to enhance the therapeutic profile of dexamethasone in rheumatoid arthritis. Although glucocorticoids such as dexamethasone (DEX) offer stronger anti-inflammatory activity than disease-modifying and biologics drugs, long term use or high dosage of glucocorticoids is associated with severe systemic adverse effects (e.g. immunosuppression, hyperglycemia, and osteopenia), due to non-specific distribution.

The authors developed a CRV-DEX conjugate, the conjugation consists a short polyethylene glycol (PEG) linker and a reactive oxygen species (ROS) responsive linker, which is mainly cleaved intracellularly, to allow the drug release specifically in inflamed tissues of where DEX is linked to a macrophage targeting peptide (CRV) through ROS, to improve stability and specific targeting.

The study found that CRV peptide selectively accumulated in inflamed synovial tissue of arthritic joints after IV injection, it showed strong binding to macrophages and neutrophils. The receptor for CRV, retinoid X receptor beta (RXRB), was significantly upregulated. Moreover, CRV-DEX significantly increased dexamethasone accumulation in inflamed joints, at the same time drug distribution to healthy organs was reduced. Additionally, it showed stronger anti-inflammatory effects compared to free DEX. It reduced synovial inflammation and cartilage damage in RA models, as well as downregulating inflammatory cytokines (e.g. IL6, IL1β, IL18, MCP-1, TNFα). Furthermore, CRV-DEX significantly decreased glucocorticoid toxicity in immune organs (thymus and spleen), reduced apoptosis in immune tissues compared to free DEX. CRV-DEX treated mice showed reduced risk of candida albicans. Finally, RXRB was elevated and co-localized with inflammation in RA patients compared to osteoarthritis control.(6)

Recently, researchers began studying the feasibility of the oral route for peptides as a replacement for subcutaneous injections as patients prefer oral medication to injectables. However, this is proving to be a challenge; as peptides undergo enzymatic degradation along the GI tract and are subjected to the rough stomach acidic conditions through the oral route. Moreover, they have to surmount the physical barriers such as the proteolytic rich intestinal mucus in order to reach systemic circulation.

Even with such hurdles, solutions for it are being explored such as structural modification like cyclization, formulation optimizations, such as micro-emulsions, absorption enhancers, enteric coatings, and the inclusion of peptidase inhibitors in the formulation.(7)

Osteoarthritis Treatments with Therapeutic Peptides

When talking about inflammatory diseases, osteoarthritis comes to mind. OA is a severe degenerative joint disease that significantly impairs the quality of life. It typically presents as articular cartilage degeneration, synovial lesions and subchondral bone sclerosis. (8) Peptide AOD-9604 was investigated in a rabbit model of osteoarthritis, which demonstrated enhancements in joint surface integrity and cartilage morphology. This peptide was even investigated as a treatment protocol for OA with hyaluronic acid or platelet-rich plasma.(9)

Conclusion

This article demonstrates the findings of the effects and uses of therapeutic peptides in inflammatory diseases such as RA and OA in different studies and trials. The results suggest that peptides enhance the delivery or the effects of existing arthritis treatments, and they themselves could be used as one. While this field of study has already showed promising results, more research, studies, and clinical trials are required to fill in the gaps to reach a wildly available tolerable treatment for patients.

References

1.

Liu C, Yan Z, Zhang X, Xia T, Ashaolu JO, Olatunji OJ, et al. Food-derived bioactive peptides potentiating therapeutic intervention in rheumatoid arthritis [Internet]. [cited 2026 Jun 28]. Available from: https://www.cell.com/heliyon/abstract/S2405-8440(24)07135-4

2.

Advance in peptide-based drug development: delivery platforms, therapeutics and vaccines - PMC [Internet]. [cited 2026 Jun 28]. Available from: https://pmc.ncbi.nlm.nih.gov/articles/PMC11880366/

3.

Bustos AH, Brüner M, Kragstrup TW, Astakhova K. Citrullinated peptides as drug candidates for rheumatoid arthritis. Front Immunol. 16:1648913. doi:10.3389/fimmu.2025.1648913 PubMed PMID: 41376634; PubMed Central PMCID: PMC12685853.

4.

Efficacy of Bioactive Cyclic Peptides in Rheumatoid Arthritis: Translation from In Vitro to In Vivo Models - PMC [Internet]. [cited 2026 Jun 28]. Available from: https://pmc.ncbi.nlm.nih.gov/articles/PMC6151468/

5.

Nanjaiah H, Moudgil KD. The Utility of Peptide Ligand-Functionalized Liposomes for Subcutaneous Drug Delivery for Arthritis Therapy. International Journal of Molecular Sciences. 2023 Apr 7;24(8):6883. doi:10.3390/ijms24086883

6.

Wu X, Guo H, Gao H, Li Y, Hu X, Kowalke MA, et al. Peptide targeting improves the delivery and therapeutic index of glucocorticoids to treat rheumatoid arthritis. J Control Release. 2024 Apr;368:329–43. doi:10.1016/j.jconrel.2024.02.040 PubMed PMID: 38431094; PubMed Central PMCID: PMC11001515.

7.

Stein Gold L, Eyerich K, Merola JF, Torres J, Coates LC, Allegretti JR. Oral Peptide Therapeutics as an Emerging Treatment Modality in Immune-Mediated Inflammatory Diseases: A Narrative Review. Adv Ther. 2025;42(7):3158–72. doi:10.1007/s12325-025-03213-8 PubMed PMID: 40439953; PubMed Central PMCID: PMC12182492.

8.

Chen B, Ning K, Sun M li, Zhang X an. Regulation and therapy, the role of JAK2/STAT3 signaling pathway in OA: a systematic review. Cell Commun Signal. 2023 Apr 3;21:67. doi:10.1186/s12964-023-01094-4 PubMed PMID: 37013568; PubMed Central PMCID: PMC10071628.

9.

Rahman OF, Lee SJ, Seeds WA. Therapeutic Peptides in Orthopaedics: Applications, Challenges, and Future Directions. J Am Acad Orthop Surg Glob Res Rev. 2026 Jan 2;10(1):e25.00236. doi:10.5435/JAAOSGlobal-D-25-00236 PubMed PMID: 41490200; PubMed Central PMCID: PMC12753158.